WORKED SOLUTION

STEP II 2001, Question 10

Mechanics • Collisions and Restitution

STEP II 2001, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/01-s2-q10

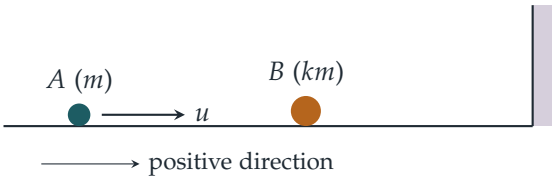
The key idea

This is two collisions and a race. The first collision is the standard pair of equations: conservation of momentum and Newton's experimental law. After B bounces off the wall, the only question is whether B, now coming back, can meet A. That happens exactly when A's velocity towards the wall is greater than B's, which turns into the inequality for k. For the energy part we write the kinetic energy as a function of k and compare it with $mu^2/27$. The comparison turns out to be a perfect square.

Solution

The second collision

Model A and B as particles on a smooth surface, so the only horizontal impulses come from the collisions. Take the direction from A towards the wall as positive. Let v_A and v_B be the velocities of A and B just after their first collision.



First collision. Momentum is conserved and the separation speed is e_1 times the approach speed:

$$\begin{aligned} mv_A + km\,v_B &= mu, \\ v_B - v_A &= e_1 u. \end{aligned}$$

Adding k times the second equation to the first gives $(1+k)v_B = (1+e_1)u$. So

$$v_B = \frac{(1+e_1)u}{1+k}, \qquad v_A = v_B - e_1 u = \frac{(1-ke_1)u}{1+k}.$$

Since $v_B > 0$, B moves on towards the wall. A may be moving forwards, at rest, or moving backwards, depending on the sign of $1-ke_1$.

The wall. B hits the wall at speed v_B and rebounds at speed $e_2 v_B$, so its velocity becomes $-e_2 v_B$. Meanwhile A keeps moving with constant velocity v_A. Note that A cannot reach the wall first, because $v_A < v_B$ (the difference is $e_1 u > 0$) and A starts behind B.

Ask yourself After the rebound A and B might be moving in the same direction. How do we decide whether they meet?

After the rebound B is between A and the wall and nothing else acts on either particle. So the gap between them closes if and only if A's velocity is greater than B's:

$$v_A > -e_2 v_B \iff v_A + e_2 v_B > 0.$$

If $v_A > 0$ they are moving towards each other. If $v_A \le 0$, B is chasing A and catches it only if it is faster. The single inequality covers both cases. Substituting, and multiplying by the positive quantity $(1+k)/u$, we get

$$1 - ke_1 + e_2(1+e_1) > 0 \iff k < \frac{1+e_2(1+e_1)}{e_1},$$

dividing by $e_1 > 0$ in the last step. So there is a second collision provided this inequality holds, as required. (If equality holds, A and B move off with the same velocity and never meet.)

The kinetic energy

With $e_1 = \frac{1}{3}$ and $e_2 = \frac{1}{2}$,

$$v_A = \frac{(3-k)u}{3(1+k)}, \qquad e_2 v_B = \frac{1}{2}\cdot\frac{\frac{4}{3}u}{1+k} = \frac{2u}{3(1+k)}.$$

Just after B rebounds, A still has speed $|v_A|$ and B has speed $e_2 v_B$. So the total kinetic energy is

$$\begin{aligned} T &= \tfrac{1}{2}mv_A^2 + \tfrac{1}{2}km\,(e_2 v_B)^2 \\ &= \frac{mu^2}{18(1+k)^2}\left[(3-k)^2 + 4k\right] = \frac{mu^2\,(k^2-2k+9)}{18(1+k)^2}. \end{aligned}$$

Now compare with $mu^2/27$, using the common denominator $54(1+k)^2$:

$$\begin{aligned} T - \frac{mu^2}{27} &= \frac{mu^2}{54(1+k)^2}\left[3(k^2-2k+9) - 2(1+k)^2\right] \\ &= \frac{mu^2}{54(1+k)^2}(k^2-10k+25) = \frac{mu^2\,(k-5)^2}{54(1+k)^2}. \end{aligned}$$

Ask yourself What is the condition $k < 5$ for, if a square is never negative?

The right-hand side is never negative, and it is zero only when $k = 5$. So for $k < 5$ we have $(k-5)^2 > 0$ and therefore

$$T > \frac{mu^2}{27},$$

as required. The value 5 is not a coincidence. With these coefficients the first part's bound is $(1 + \frac{1}{2} \cdot \frac{4}{3})/\frac{1}{3} = 5$. So $k < 5$ is exactly the condition for a second collision. At $k = 5$ both particles move at $-u/9$ after the rebound, and the energy is exactly $mu^2/27$.

Checks

- Take $e_1 = 1$ and $k = 1$ (equal masses, perfectly elastic). Then $v_A = 0$ and $v_B = u$, the familiar exchange of velocities. The bound becomes $k < 1 + 2e_2$, which $k = 1$ satisfies whenever $e_2 > 0$: a returning B will certainly hit a stationary A.
- For $k = 5$ the formula gives $v_A = -\frac{u}{9}$ and $-e_2 v_B = -\frac{1}{2} \cdot \frac{2u}{9} = -\frac{u}{9}$, equal velocities, as claimed.
- Numerically, with exact fractions for k from 0.001 to 4.999, the ratio $T/(mu^2/27)$ has minimum 1.00000001 near $k = 5$ and is always above 1. A random test of 20 000 triples (k, e_1, e_2) found no case where the inequality and a direct comparison of the post-rebound velocities disagree.

Common slips

- Requiring $v_A > 0$. A can be moving backwards and still be caught by B. The correct condition compares the two velocities after the rebound.
- Forgetting that the wall reverses B's direction. Its velocity becomes $-e_2 v_B$, not $e_2 v_B$.
- Using B's speed before the wall in the energy. The question wants the energy just after the rebound, so B's speed is $e_2 v_B$.

Beyond STEP

Newton's experimental law is a model for what happens in the short, violent deformation during an impact. The coefficient of restitution summarises how much of the relative kinetic energy is lost. In a first-year mechanics course, collisions are often analysed in the centre-of-mass frame. In that frame a collision simply reverses the relative velocity and multiplies it by e, and the energy lost is $\frac{1}{2}\mu(1-e^2)w^2$, where μ is the reduced mass and w the approach speed. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops these ideas carefully.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.